Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Multi-Modal Time-Series Diagnostics for Metal Additive Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	Materials Analysis/Evaluation Technologies	Robotics/Automation Machinery	Measurement/Sensor Technologies for Automation
Concept of Proposed Technology	Metal additive manufacturing (AM) is moving from prototyping to production, but for many small and medium-sized enterprises (SMEs) the limiting factor is not the printer itself, but rather the **cost, uncertainty, and throughput penalty** associated with quality verification. Market signals strongly reinforce this: a recent assessment summarized that the broader quality control/inspection and testing ecosystem is measured in the hundreds of billions of dollars, while AM revenues are forecast to grow strongly through the decade (e.g., product testing/inspection/certification projected from $214B in 2020 to $349B by 2030) [1-3]. At the same time, AM market projections consistently indicate rapid expansion: estimates span from roughly $15B in 2023 to $34.5B by 2028 (18.1% CAGR) to $20.37B in 2023 to $88.28B by 2030 (23.3% CAGR). Independent industry tracking also reports continued growth and increasing metal-system sales; for example, Wohlers Report 2024 (via ASTM) cites $20.035B AM industry revenue in 2023 and 3,793 metal AM systems sold in 2023. The implication for SMEs is clear: as AM penetrates production supply chains, the competitive advantage shifts to those who can produce **certifiable parts with high first-pass yield** and minimal post-process inspection burden. The combination of fast AM growth with an inspection ecosystem creates a clear commercialization target: **in-process, real-time defect determination that reduces unnecessary post-build inspection and prevents wasted builds.**			

For SMEs, the objective is to make **timely and accurate decisions** that reduce scrap/rework and focus inspection resources where they add value. In practice, even modest improvements in (i) early detection of "doomed builds," and (ii) reduced operator trial-and-error can materially improve cost per qualified part. The proposed technology is designed around this concept: convert streaming, heterogeneous process data into three **actionable outcomes**, i.e., **continue**, **intervene**, or **inspect**, each accompanied by a **quantified confidence level** and an **interpretable diagnostic rationale**. This aligns with broader national/industry positioning that in-situ monitoring is key to lowering inspection cost and improving qualification strategies for metal additive manufacturing processes. The proposed technology addresses this bottleneck by delivering a **retrofit-friendly, data-efficient, and computationally lightweight** in-process diagnostics capability for metal AM. We have already tested this approach on **laser powder bed fusion (LPBF)** and are planning on extending the sensing/analytics concepts and validation protocols to **wire-arc additive manufacturing (WAAM)**. The core idea is to use heterogeneous streaming signals (such as melt pool images, acoustic emissions, etc.) for real-time fault detection and then decision trees associated with detection. This is essential for SMEs that often cannot afford extensive labeled datasets of rare anomalies, nor maintain complex cloud infrastructures.

Multi-modal sensing designed for SME integration

The proposed system fuses **multi-rate, multi-modal time series** into a coherent "quality state" estimate: (i) **acoustic emission (AE) / acoustic sensing** and **structure-borne vibration** (high bandwidth), (ii) machine "digital thread" telemetry such as **laser power, scan speed, gas flow**, and, for WAAM, **wire feed and current/voltage**, and (iii) optional **thermal/vision signatures** when available. This architecture deliberately prioritizes **acoustic/structure-borne sensing** because it offers **low integration cost** and **high temporal resolution**, and because published work indicates that both imaging and acoustic modalities are relevant to in-situ detection of subsurface/internal defect formation and process instability.

This project integrates **guided-wave/structure-borne acoustic sensing through the build plate** using piezoelectric elements that can function as both actuators and sensors. This enables two complementary monitoring modes:

1. **Passive listening** (AE/vibration signatures associated with abnormal melt-pool dynamics, recoater interactions, spatter-driven disturbances, or unstable deposition regimes).
2. **Active interrogation** (periodic guided-wave probes whose propagation metrics change with evolving porosity, lack-of-fusion, and geometric/track anomalies).

This "passive + active" architecture is intended to strengthen sensitivity to defect formation mechanisms that are not reliably captured by cameras alone, while remaining implementable on SME-relevant machines. This aligns with the market assessment's emphasis that acoustic monitoring can be deployed as either passive or active sensing and can be sensitive to pores/cracks/un-melted material that alter wave transmission.

Modular, integrated system architecture

To reduce adoption friction, the technology is structured as a **modular stack** that can be deployed incrementally:

- **Sensing layer:** multi-rate, multi-modal inputs including AE/structure-borne vibration (high bandwidth), machine "digital thread" data stream (laser power, scan speed, gas flow; and for WAAM, wire feed/current/voltage), plus optional thermal/vision when available (Summary.pdf).
- **Edge analytics layer:** computationally lightweight streaming algorithms that run near-real time, producing anomaly scores and probabilistic thresholds without requiring large labeled anomaly libraries.
- **Decision and reporting layer:** an operator dashboard/API that logs audit-ready evidence (time-stamped signals/features, detected events, confidence bounds) and generates "continue/intervene/inspect" guidance.
- **Optional feedback pathway:** interfaces for machine parameter adjustment or operator interventions when SMEs choose to progress toward closed-loop control.

This architecture directly addresses the central SME need articulated in the Summary: **retrofit-friendly**, **data-efficient**, **computationally lightweight**, and **actionable** monitoring.

Hybrid time-series analytics with physics- and process-informed structure

The proposed data analytics approach combines **streaming statistical monitoring** (anomaly detection, fault/defect identification and/or classification, and probabilistic thresholds) with **physics- and process-informed constraints** to reduce false alarms under changing operating conditions (toolpaths, geometries, thermal histories). Concretely, the system builds baseline models of "normal" behavior in the relevant signal subspaces and then detects departures using interpretable diagnostics while explicitly accounting for the fact that LPBF/WAAM signals are **nonstationary** and operating regimes shift during a build.

Because SME deployments must operate near-real-time, the approach emphasizes **edge-computable features and models**: multi-rate alignment of telemetry and vibro-acoustic streams; time-frequency descriptors for acoustic signatures; and regime-aware baselines conditioned on process parameters. When anomalies are detected, the system performs a second-stage **diagnostics inference** to separate classes of anomalies/faults that matter operationally (e.g., **void-related** versus **geometric/track** anomalies initially), providing the operator with likely cause categories and recommended actions. This approach directly supports SME workflows as it enables rapid intervention when needed and supports **inspection triage** thus reducing unnecessary CT scans or destructive tests.

TRL maturation and KPIs (3-year, TRL 3-4 start)

Starting from TRL 3-4, the project targets a clear industrialization pathway: laboratory validation to relevant-environment demonstrations with SME integration. Performance will be assessed using:

- **Detection:** missed-detection rate and false-alarm rate for prioritized anomaly classes.
- **Diagnostics:** accuracy of anomaly class identification (e.g., void-related vs. geometric/track).
- **Latency:** time from event/fault occurrence to decision output.
- **Robustness:** performance across parameter changes, geometries, and build stages along with quantified uncertainty and statistical decision-making.

Early-stage results and proof-of-concept

Early-stage results from the proposal team establish a proof-of-concept for **online, low-training-burden anomaly detection** in LPBF and for **active acoustic interrogation**. First, in a recent series of papers by the proposal team [4-7] we report feasibility studies on an open-source LPBF testbed at RPI using Inconel 718, where infrared melt-pool imaging is acquired at **2 kHz** and compressed into physically interpretable scalar features. The studies evaluate multiple unsupervised statistical features across **various raster geometries** and multiple anomaly scenarios (e.g., edge anomalies, injected "laser-off" squares to emulate lack-of-fusion, and naturally occurring undermelt/spatter events). Notably, the best-performing methods achieved **very low false-alarm rates of 0.49% and 0.37%**, respectively, while remaining compatible with online deployment and sparse nominal training. Equally important for industrial translation, the approach supports **spatial localization** of anomalous signatures by mapping time-indexed features back to the known scan path.

Second, **a submitted patent** by the proposal team presents the integration of **guided-wave acoustic sensing** into the RPI AM chamber, including a build-plate sensor/actuator array and vacuum-compatible wiring/DAQ integration demonstrating a practical pathway to chamber-ready acoustic instrumentation. In preliminary experiments, guided waves were transmitted through a build plate and quantified via a "damage index." Using multiple sensor pairs and frequencies, the method **clearly differentiated damage/anomalies** across repeated trials (20 runs per condition), indicating sensitivity to incremental material addition or geometric discontinuities. Together, these results justify a hybrid passive/active sensing roadmap: camera-derived features enable rapid online screening, while guided waves provide an extensible route toward **structure-borne and potentially subsurface-sensitive** diagnostics.

Rationale for Joint R&D Project

A joint R&D project is justified for the following reasons:

- **Business value: higher yield and faster throughput**
  The joint R&D effort targets measurable reductions in scrap, rework, and rebuilds by detecting off-nominal conditions early and enabling timely intervention. For an SME, this directly improves machine utilization and lowers cost per qualified part, creating a clear ROI pathway that can be validated on representative production builds.

	• **Reduced inspection burden with audit-ready evidence** Rather than treating post-build CT or destructive testing as the default, the project will develop "inspect only when warranted" decision logic supported by traceable data logs and confidence metrics. This enables risk-informed inspection planning, shortens lead times, and provides documentation suitable for customer audits and internal quality systems. • **Workflow-fit deployment in a relevant environment (TRL 3-4 maturation)** The SME partner provides the operational context needed to move from laboratory feasibility to a deployable solution: machine interfaces, material variability, operator practices, and production constraints. Joint development will produce practical integration installation, calibration, and maintenance procedures so the technology fits the SME rather than remaining a lab prototype. • **Differentiated multi-modal sensing and decision support** The project co-develops a robust monitoring methodology that combines guided-wave acoustics/vibration with machine data streams (and optional imaging), providing resilience to optical artifacts and expanding sensitivity to process/part anomalies. Importantly, the output is not raw monitoring data but actionable recommendations, *i.e., continue, intervene, or inspect,* with quantified confidence and interpretable rationale. • **Representative data, defect modes, and generalization** Access to real parts, geometries, and defect drivers (powder/wire lots, machine drift, parameter changes) is essential for building models that generalize beyond curated experiments. The SME environment supplies these realities, allowing the team to prioritize the highest-value anomaly classes and validate robustness across normal production variability. • **Shared risk, faster transfer, and competitive positioning** The university team de-risks instrumentation and analytics development, while the SME ensures relevance and accelerates time-to-value through iterative trials. With a patent already filed by RPI and the proposal team, the joint work can strengthen partner-specific implementations and support a defensible differentiation strategy in the SME's target markets.
Key Words	1. Real-time anomaly detection 2. In-process quality assurance 3. Additive manufacturing 4. Inspection cost reduction 5. Laser bed powder fusion

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	The proposed R&D will mature RPI's patent-pending approach into an SME-relevant pipeline by pursuing the following objectives: 1. **Develop a retrofit-ready, multi-modal sensing package for LPBF (with WAAM extensibility).** Design and validate an instrumentation topology that fuses structure-borne acoustics/vibration (including guided-wave-capable piezoelectric elements), sensor measurements (e.g., laser power/scan speed/gas flow), and optional imaging/thermal inputs when available. Emphasis is placed on practical mounting, chamber compatibility, and repeatable calibration. 2. **Create edge-executable, streaming anomaly detection for nonstationary AM time series.** Implement computationally lightweight time-series monitoring algorithms that operate in near-real time under changing toolpaths, geometries, and thermal states. Deliver robust anomaly/fault detection with statistical thresholds and quantified confidence, minimizing reliance on large, labeled anomaly datasets. 3. **Develop diagnostic inference that maps anomalies to actionable cause categories.** Extend detection to diagnosis by distinguishing high-value anomaly classes (initially void-related and geometric/track anomalies, then expanding to additional defect families). Provide interpretable evidence (features, time-frequency signatures, regime context) that			

	supports root-cause analysis. 4. **Implement decision support outputs aligned with SME workflows.** Translate streaming analytics into operational recommendations, i.e., continue / intervene / inspect, with confidence bounds, alerting protocols, and audit-ready logs. Ensure outputs integrate with existing quality processes and support targeted post-build inspection rather than destructive testing. 5. **Validate performance and robustness on representative builds and variability sources.** Demonstrate repeatable performance across representative parts, parameter ranges, and variability sources (materials, machine drift, operator effects). Quantify KPIs such as false-alarm rate, missed-detection rate, decision latency, and robustness across build stages. 6. **Deliver integration protocol for adoption and scale-up.** Provide an SME-ready package including installation/calibration SOPs, data interfaces (dashboard/API), documentation for traceability, and a roadmap for platform expansion (LPBF-first, WAAM extension) and future closed-loop capabilities.
R&D Target Specifications	• **Anomaly detection performance:** Event-level false-alarm rate ≤ 2%. Probability of detection (POD) ≥ 90% for actionable anomalies. Detection timeliness: ≤ 1 layer from onset (LPBF) or ≤ 1–3 s (WAAM). • **Anomaly localization performance:** Localize anomalies to ≤ 2-5 mm in XY and ≤ 1-2 layers in Z (LPBF), reported as affected-region tiles/polygons. ≥ 99% of flagged events time-stamped and mapped to toolpath coordinates, layer number, and build metadata. • **Decision-support outcomes:** Provide operator-ready recommendations, i.e., continue / intervene / inspect, with confidence bounds. Target ≥ 90% agreement with post-build ground truth for decisions after tuning on representative builds. • **Real-time edge execution:** End-to-end latency ≤ 1 s from acquisition to decision output (aspirational ≤ 250 ms for high-rate channels). Deployable on an industrial edge PC without GPU; typical footprint 8-16 CPU cores and ≤ 8 GB RAM for streaming inference and logging.
R&D Plan	This 3-year joint R&D program will mature a retrofit-ready, multi-modal in-process monitoring and decision-support capability for metal AM, with an LPBF implementation and a clear extension path to WAAM. The plan is organized into five tasks that progress from information compression and time-series generation and monitoring, to guided-wave acousto-ultrasonics for build-plate interrogation, to multi-modal fusion and operational decision logic, and finally to a "Build Supervisor" AI agent and rigorous validation in representative SME conditions. Across all tasks, the emphasis is on deployable engineering artifacts, i.e., sensor integration guidance, edge-executable analytics, dashboard/API outputs, SOPs, and audit-ready reporting, alongside measurable performance targets (false alarms, detection probability, localization, latency, and usability). **Task 1 — Melt-Pool Signal Compression and Time-Series Diagnostics** **Objective.** This task develops the LPBF imaging-to-time-series pathway and the associated streaming diagnostics capability that converts high-rate melt-pool monitoring into actionable, traceable anomaly/fault/defect detection, identification, and localization events suitable for targeted inspection and operator intervention. **Approach and outputs.** The starting point is the practical observation that high-rate melt-pool images are information-rich but operationally burdensome: storing and interpreting full image streams is costly and rarely necessary for shop-floor decisions. The approach therefore focuses on compressing the image stream into a small set of physically interpretable scalar signals, such as melt-pool size/area and basic intensity/stability measures, that retain sensitivity to process parameters while dramatically reducing data volume and computational load. These compressed signals form time series aligned to the scan path and build context (layer, region, scan strategy and raster pattern), enabling traceability from an alert back to "where on the part" and "under what process conditions" it occurred. On top of these time series, the task implements data-efficient advanced stochastic modeling to establish a baseline of expected behavior and quantify deviations in real time. The goal is not simply to flag points that look "different," but to form actionable events with severity and duration, control alarms, and generate outputs that can be used to guide targeted inspection and corrective action. The engineering deliverables from this task include a reusable feature-extraction module (camera-agnostic where possible), an edge-executable detector with tunable thresholds to meet project false-alarm targets, and an event-to-toolpath mapping capability that outputs affected-region(s) and anomaly locations for downstream inspection planning. **Milestones.** By the end of Year 1, the image-to-time-series pipeline and an online baseline detector will run on an industrial edge PC, producing time-stamped events. During Year 2, anomaly/event

localization will be refined and standardized into affected-region outputs suitable for anomaly identification and localization. Also in Year 2, the baseline model set will be expanded to account for broader variability (parameter windows, geometries, and build stages) without excessive retraining. Emphasis will be placed on implementing geometry generalization capabilities so that the model can be used on any build geometry with minimal retraining.

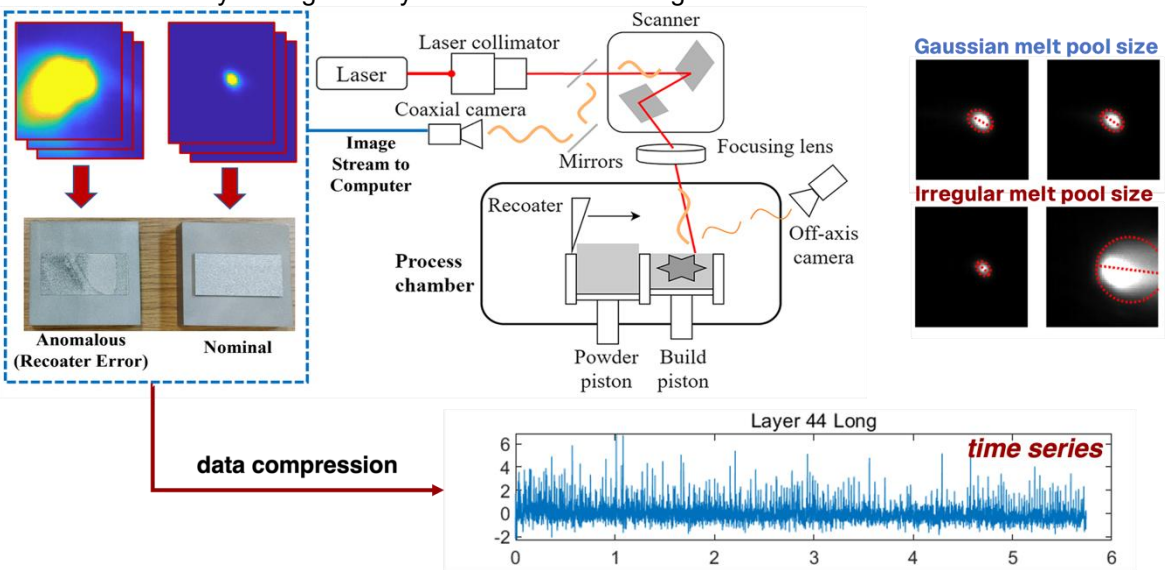

Figure 1. Flowchart of the proposed data compression and time series generation method [4].

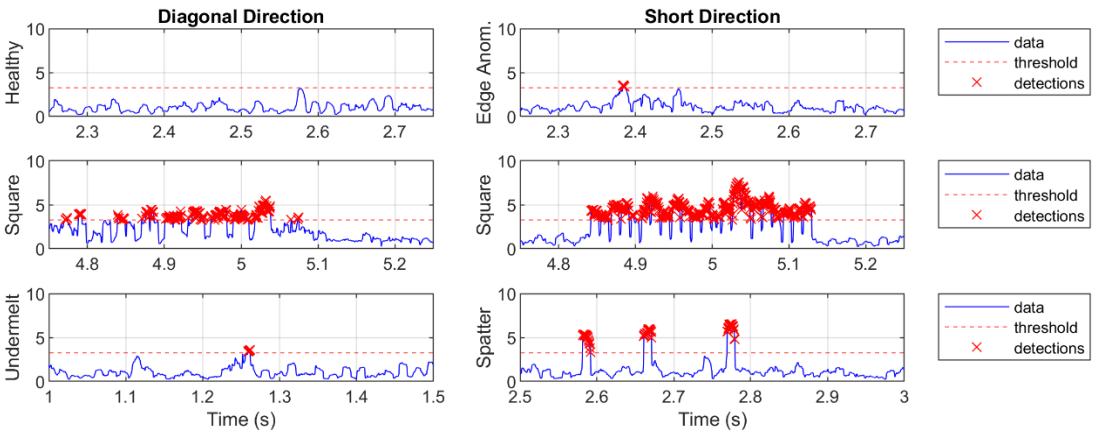

Figure 2. Indicative experimental anomaly detection results. Detections are labeled with a red "x," the detection threshold is illustrated with a red, dashed line, and the detection statistic All anomalies are accurately detected, and the nominal dataset has no false alarms [4, 5].

Task 2 — Acousto-Ultrasonic Guided-Wave Sensing for Build-Plate Interrogation

Objective. This task develops and demonstrates an active acousto-ultrasonic guided-wave sensing capability integrated with the build plate to provide a complementary monitoring channel that is sensitive to minor build defects, robust to optical, i.e., melt pool, artifacts and, over time, supports sensitivity to structure-borne changes associated with build evolution and defect formation.

Approach and outputs. Optical/melt-pool sensing is valuable but can be affected by line-of-sight limitations, plume/spatter interference, and challenges in connecting surface images and signatures to subsurface integrity. Guided-wave sensing addresses these limitations by injecting controlled wave packets through the build plate (and, as appropriate, through evolving build conditions) and measuring how the received waveforms change across sensor paths. The task will implement a practical transducer topology using piezoelectric elements that can function as both actuators and sensors, with integration choices driven by chamber compatibility, wiring/DAQ specifications and limitations, and retrofit potential. Based on physics-based models, simulations, and wave mechanics, the development will focus on robust, repeatable, and defect/fault sensitive measurements as well as on the extraction of "health indices" from received signals, such as difference metrics, energy measures, phase/dispersion evolution and consistency, and related features that can be computed reliably on an edge device. Early experimentation will investigate

use controlled defects such as known mass or stiffness perturbations and build-plate condition variations, to quantify repeatability, sensitivity, and robustness before progressing to in-process trials coordinated with build states (e.g., controlled scan segments). The outputs include a chamber-ready guided-wave sensing prototype, automated actuation/measurement scripts, real-time feature computation, and sensor-health diagnostics with quantified uncertainty and confidence intervals.

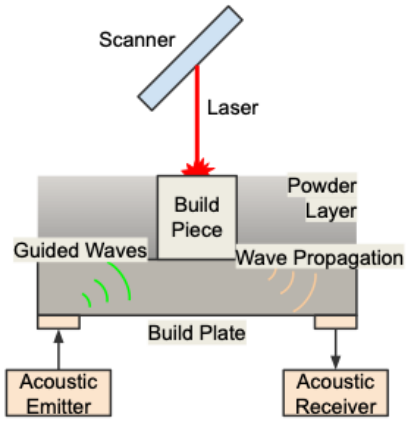

Figure 3. Schematic of the build-plate integrated ultrasonic wave sensing.

Milestones. In Year 1, the guided-wave hardware integration and repeatability characterization will be completed, establishing that the sensing layer can run reliably in the intended environment. In Year 2, sensitivity to controlled defects/faults/anomalies relevant to AM build evolution will be investigates and quantified, and the most robust "health indices" selected. In Year 2, in-build guided-wave trials will be demonstrated and integrated into the broader monitoring timeline to support fusion in Task 3.

Task 3 — Multi-Modal Fusion and "Continue / Intervene / Inspect" Decision Protocols

Objective. This task converts multiple data streams, i.e., imaging-derived time series, guided-wave features, and AM machine data streams, into a unified decision-support probabilistic output aligned with SME workflows.

Approach and outputs. SMEs need decisions they can act on, not merely more data. The fusion strategy therefore begins with the postulation of a time alignment method across sensors with different sampling rates and latencies, and with explicit conditioning on process context such as scan regime, layer, region, and parameter set. The analytics are structured as a two-stage inference pipeline: a fast-screening layer that provides **low-latency anomaly detection**, followed by a **diagnostic refinement layer** that assigns likely cause categories and ranks event severity. This design is intended to keep edge compute requirements low while improving diagnostic outputs when an anomaly is suspected. The outcome is a triage recommendation, namely **continue, intervene, or inspect**, paired with confidence bounds and a short, interpretable rationale that cites the evidence and the affected region. The task also delivers the "audit-ready evidence package" that SMEs need: time-stamped event records, supporting features and confidence metrics, and the associated build metadata. Engineering outputs include fusion algorithms that remain stable even when one sensing channel is degraded or temporarily unavailable, alarm management that limits false alarms and groups events into actionable units, and standardized data interfaces (dashboard and API outputs) that fit common SME data practices and enable downstream reporting.
Milestones. During Year 2, the first integrated multi-modal prototype will produce unified event logs and triage outputs. Over Year 2-3, fusion and calibration will reduce alarm burden while improving diagnostic accuracy and robustness under variability. In Year 3, the decision-support stack will be release-ready for SME pilot deployment, supported by KPI evidence generated under Task 5.

Task 4 — Build Inspection Supervisor AI Agent for Operator Guidance and Assessment

Objective. This task develops an operator-facing "Build Inspection Supervisor" layer that facilitates adoption and autonomous operation by explaining alerts, suggesting next actions under defined guardrails, and automating build reporting and knowledge capture.

Approach and outputs. Even when analytics perform well, adoption can fail if shop-floor users do not trust or understand the system. The Build Inspection Supervisor is therefore designed as a constrained, evidence-grounded assistant that utilizes the outputs from Task 3, event metadata, confidence metrics, process context, and prior-case summaries, and produces structured, auditable outputs. These outputs include a recommended action (continue/intervene/inspect), a concise rationale that references the specific evidence behind the decision, and an automatically generated build-quality summary that can be used for internal review and customer communication. Over time, the Supervisor's reports become a mechanism for knowledge transfer: they record what happened, where, what action was taken, and what outcome followed, helping SMEs reduce dependence on individual operator experience and accelerating process learning. Deliverables include report templates aligned with typical SME quality workflows, a lightweight deployment architecture, and integration into the dashboard/API outputs.
Milestones. In Year 2, an initial Build Inspection Supervisor prototype will generate structured

	explanations and reports linked directly to event evidence. In Year 3, the Supervisor will be integrated into the pilot workflow and refined in collaboration with the SME partner to match quality-system documentation needs and preferred operator interaction patterns. **Task 5 — Experimental Validation, Ground Truth, and SME Pilot Deployment** **Objective.** This task provides systematic performance validation and assessment, and ensures the system is deployable in SME-relevant environments by delivering acceptance-tested Standard Operating Procedures (SOP), calibration routines, and key performance indicator (KPI) evidence tied to ground truth. **Approach and outputs.** The task begins by defining a staged build matrix that includes representative nominal variability and controlled anomalies/defects where feasible, complemented by naturally occurring anomalies. Ground truth is established through targeted NDE and/or metallography focused on flagged regions so that detection performance can be measured without requiring exhaustive inspection of every build. This ground-truth strategy enables a statistically meaningful evaluation and assessment of probability of detection (POD), false-alarm rates, localization accuracy, latency, and triage correctness. Robustness is assessed explicitly across variability parameters that matter in practice, including material lots, parameter windows, geometry families, machine drift, and build stages. The task also addresses a pilot deployment either at the SME or RPI site, or on SME-representative hardware where the system is installed, calibrated, and run under acceptance criteria, producing final SOPs for installation, calibration verification, sensor-health outputs, and report generation. Outputs include validation datasets, KPI reports, acceptance test procedures, and a deployment-ready package that is credible for both internal decision-making and customer-facing qualification discussions. **Milestones.** By the end of Year 1, the validation protocol and ground-truth pipeline will be established and baseline KPIs reported. In Year 2, performance will be quantified across multiple geometries and parameter sets, enabling refinement of thresholds and alarm policies to meet target specifications. In Year 3, the SME pilot will be completed with acceptance testing, finalized SOPs, and the consolidated evidence package supporting adoption and scale-up. **Integrated Timeline and Program Milestones (3 Years)** **Year 1 (Foundation and prototypes).** The program will deliver the first deployable components: Task 1 will produce the image-to-time-series pipeline and baseline online anomaly detection; Task 2 will complete guided-wave sensor integration and uncertainty/robustness characterization; and Task 5 will establish the validation matrix and ground-truth workflow, yielding the first KPI baseline. **Year 2 (Integration and decision outputs).** Task 3 will deliver multi-modal fusion and the initial "continue/intervene/inspect" decision logic with confidence bounds. Task 4 will deliver the first Build Inspection Supervisor capability for explanations and structured reporting. Task 5 will expand validation across representative builds and variability sources, refining thresholds to meet false-alarm and timeliness targets. **Year 3 (Pilot and deployable package).** Tasks 3 and 4 will converge into a hardened edge software stack with dashboard/API outputs and audit-ready logs. Task 5 will execute the SME pilot deployment and acceptance testing and finalize SOPs and documentation. The program will close with a deployment-ready kit and a roadmap for platform expansion (LPBF-first, WAAM extension), positioned for technology transfer in alignment with the team's IP strategy.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The broad scope of the proposed effort requires a diverse team of experts spanning multiple domains. **Prof. Fotios Kopsaftopoulos** will contribute to the sensing, modeling, anomaly detection, structural health monitoring and structural mechanics, as well as computational modeling of embedded sensing systems. **Prof. Sandipan Mishra** specializes in AM process monitoring and control and will lead the experimental and testing strategies for LBPF and WAAM processes. The research team will be supported by two Ph.D. students and two undergraduate researchers who will carry out experimental, modeling, and integration tasks. RPI's **Center for Smart Convergent Manufacturing Systems (CSCMS)** provides all required manufacturing capabilities including a LPBF system, robotic WAAM cell with coordinated multi robot operation, a FormAlloy system for rapid alloy development, an AFP platform, and a Mikrotools hybrid micro machining center, along with a robust convergent manufacturing control architecture for real time sensing and

	tool tip based correction. Additional resources include Prof. Kopsaftopoulos' **Intelligent Structural Systems Lab**, equipped with a 100 kN dynamic testing frame, environmental chamber, NI data acquisition systems, and a Dolphicam ultrasound imaging system, as well as the RPI's **Materials Research Center (MRC)** for high temperature mechanical and microstructural characterization. Together, these facilities provide comprehensive support for the proposed research and ensure seamless collaboration with **Korean partners**.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to be a metal-AM part producer or service provider with access to commercial LPBF operations, production constraints, and quality-inspection workflows (e.g., printing metal components, AM prototyping and manufacturing implementation services, semiconductor equipment parts, dental implants, etc.). The Korean SME partner will play a central role in translating RPI's **patent-pending** sensing/analytics technology into an industry-relevant monitoring capability that can be deployed on production equipment and evaluated against practical quality outcomes. In particular, the SME will: • **Define industrial requirements and priority use cases** by providing representative part families, acceptance criteria, operating envelopes, and production constraints (e.g., volume production of implants or functional components with established inspection practices). • **Provide access to machines, builds, and operational data** by allocating printer time and facilitating collection of in-process signals, machine telemetry, and associated build metadata on commercial systems typical of Korean manufacturing environments. • **Lead application-driven validation and benchmarking** by supplying inspection/quality outcomes (e.g., in-house quality inspection results and defect dispositions) and aligning demonstrations with internal qualification protocols and customer expectations. • **Support pilot integration and manufacturability** by assisting with practical implementation steps, sensor mounting constraints, shop-floor installation, operator workflows, and (where applicable) machine installation/training pathways that accelerate TRL progression beyond laboratory demonstrations. • **Contribute resources and collaboration bandwidth** by designating technical points of contact, participating in joint test planning and quarterly technical reviews, enabling site-based trials, and co-developing SOPs, acceptance tests, and data-sharing arrangements. • **Enable commercialization pathways** by advising on adoption barriers, cost/throughput constraints, scalability expectations in Korea, and partner-relevant field trials; engaging with the RPI team on IP positioning consistent with the already-filed patent application. Through these contributions, the SME partner ensures the R&D advances from proof-of-concept to application-relevant prototypes validated under realistic production conditions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The research is anticipated to deliver significant scientific, technological, and industrial advancements in AM. The project will deliver outcomes with the following anticipated impacts: • **Deployable in-process monitoring capability for metal AM (LPBF-first, WAAM-ready):** A retrofit-oriented sensing and edge-analytics stack that converts multi-modal in-process signals into actionable decisions (continue / intervene / inspect) with quantified confidence, event localization, and audit-ready traceability suitable for SME production workflows. • **Demonstrated performance against target specifications:** Quantified validation evidence, including controlled false-alarm rates, high probability of detection for actionable anomalies, millimeter-scale localization tied to toolpath coordinates, and real-time edge execution, supported by targeted NDE/metallography to establish credible ground truth. • **Reduced cost and lead time for qualification and inspection:** A risk-informed inspection workflow that prioritizes post-build NDE only when warranted, reducing reliance on blanket CT/destructive testing, shortening iteration cycles, and improving first-pass yield and machine utilization for the SME partner. • **Validated guided-wave acousto-ultrasonic sensing pathway:** Chamber-compatible build-plate guided-wave instrumentation with accurate and robust health indices, providing robustness under optical/melt pool artifacts and a scalable route to structure-borne (and potentially subsurface-sensitive) diagnostics that complements melt-pool imaging. • **Industrial-ready integration artifacts:** A partner-tested package including installation and calibration SOPs, sensor-health checks, acceptance tests, standardized data interfaces (dashboard/API), and automated build-quality reports reducing adoption effort and enabling replication across machines and part families. • **Commercialization and competitive advantage enabled by IP:** With RPI and the proposal team's patent-pending technology, the project is positioned to support partner-

	relevant licensing/field trials and offerings (e.g., "quality-assured" AM, monitoring-as-a-service, or value-added inspection triage) within Korean and global markets. • **Broader industrial and economic impact:** Strengthening SME competitiveness by improving production reliability and lowering the barrier to qualified metal AM manufacturing; accelerating adoption of smart manufacturing practices; contributing to a resilient supply chain by enabling repeatable, data-backed quality assurance for critical metal components.

6. References

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(4) Chen A., Kopsaftopoulos F., Mishra S., "Feasibility Study for Online Anomaly Detection in Metal Additive Manufacturing Using Statistical Time Series Methods,'' accepted, Structural Health Monitoring

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(7) Chen A., Mishra S., Kopsaftopoulos F., "Online Fault Detection for Metal Additive Manufacturing with Data-Driven Time Series Models," *Dynamic Data Driven Applications Systems (DDDAS) – InfoSymbiotics and AI*, New Brunswick, NJ, USA, November 2024. https://doi.org/10.1007/978-3-031-94895-4_2

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization